

C908

Integrated Physical Sciences

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A delivery van from a company with a fleet of 40 electric vehicles speeds up from 0 to 24 m/s in 8 s on a straight road. Its battery holds 75 kWh. What is its average acceleration?

- A. 24 m/s²
- B. About 9.4 m/s²
- C. 192 m/s²
- D. 3 m/s²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

On an air track in a lab that recently replaced its blower, a 2-kg cart moving at 6 m/s hits a 4-kg cart at rest, and they stick together. The carts are painted red and blue. How fast do they move afterward?

- A. 6 m/s
- B. 2 m/s
- C. 3 m/s
- D. 12 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A 1-kg cart hits a 4-kg cart at rest; they stick and move off together at 4 m/s. How fast was the 1-kg cart moving before the collision?

- A. 20 m/s
- B. 0.8 m/s
- C. 4 m/s
- D. 16 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A mover from a company with five-star reviews carries a 30-kg box (weight 294 N) straight across a 10-m room at constant height. How much work does the mover's upward holding force do on the box?

- A. 300 J
- B. 2,940 J
- C. Positive work, because the mover gets tired
- D. 0 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

5.

A jazz radio station broadcasts at 100 MHz (1.0×10^8 Hz) from a 150-m tower. Radio waves travel at 3.0×10^8 m/s. What is the wavelength?

- A. 150 m
- B. 3×10^8 m
- C. 3 m
- D. 3×10^{16} m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

6. [TRANSFER — reversed stem / different variable]

A wave on a rope has a wavelength of 0.5 m and a speed of 2 m/s. What is its frequency?

- A. 4 Hz
- B. 2 Hz
- C. 0.25 Hz
- D. 1 Hz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

7.

A string of warm-white LED lights sold by a hardware store with free delivery has a total resistance of 4Ω and runs on a 12-V battery. What current flows?

- A. 12 A
- B. About 0.33 A
- C. 48 A
- D. 3 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

8. [TRANSFER — reversed stem / different variable]

A 9-V battery drives 0.5 A through a small motor. What is the motor's resistance?

- A.** About 0.056 Ω
- B.** 4.5 Ω
- C.** 18 Ω
- D.** 9 Ω

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A 200-g mug of water in a staff-room microwave rated at 1,100 W is heated from 10°C to 35°C. Water's specific heat is 4.18 J/g·°C. How much heat does the water absorb?

- A.** 29,260 J
- B.** 836 J
- C.** 20,900 J
- D.** 1,100 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Sodium, a metal, and chlorine, a nonmetal, combine to make table salt, which a food company sells in 1-kg bags. What kind of bond forms, and how?

- A.** Ionic; chlorine transfers an electron to sodium.
- B.** Ionic; sodium transfers an electron to chlorine.
- C.** Covalent; the atoms share electrons.
- D.** Metallic; electrons form a shared sea.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Pond water near a closed paper mill tests at pH 5. Rainwater collected the same day at 18°C tests at pH 3. How does the rainwater's H^+ concentration compare with the pond's?

- A.** 2 times greater
- B.** 100 times greater
- C.** About 1.7 times greater
- D.** 100 times smaller

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Solution A has 1,000 times more H^+ than solution B, which has pH 9. What is the pH of solution A?

- A.** 6
- B.** It cannot go below 7 from a basic starting point.
- C.** 3
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A delivery van ~~from a company with a fleet of 40 electric vehicles~~ speeds up from 0 to 24 m/s in 8 s on a straight road. ~~Its battery holds 75 kWh.~~ What is its average acceleration?

Target word: acceleration **Decides:** $\Delta v = 24$ m/s over 8 s.

A	Dropped element	Drops the time; that is the change in speed.
B	Camouflage	$75 \div 8$ uses the battery figure.
C	Inverted operation	Multiplies by time instead of dividing.
D	Correct	$24 \div 8$.

2. Answer B Concept 2

On an air track ~~in a lab that recently replaced its blower~~, a 2-kg cart moving at 6 m/s hits a 4-kg cart at rest, and they stick together. ~~The carts are painted red and blue.~~ How fast do they move afterward?

Target word: final speed **Decides:** Momentum $2 \times 6 = 12$ kg·m/s shared by 6 kg.

A	Confusion set	Conserves speed instead of momentum.
B	Correct	$12 \div 6$.
C	Dropped element	Divides by 4 kg, leaving out the first cart's mass.
D	Adjacent category	That is the momentum in kg·m/s, not a speed.

3. Answer A Concept 2 · transfer

A 1-kg cart hits a 4-kg cart at rest; they stick and move off together at 4 m/s. How fast was the 1-kg cart moving before the collision?

Target word: solve for initial speed **Decides:** Final momentum $5 \times 4 = 20$ kg·m/s.

A	Correct	$20 \div 1$ kg.
B	Inverted operation	Divides 4 by 5 instead of multiplying.
C	Confusion set	Assumes speed is conserved.
D	Dropped element	Uses only the 4-kg mass after the collision.

4. Answer D Concept 3

A mover ~~from a company with five star reviews~~ carries a 30-kg box (weight 294 N) straight across a 10-m room at constant height. How much work does the mover's upward holding force do on the box?

Target word: work by a vertical force during level motion **Decides:** Force is vertical; motion is horizontal.

A	Wrong stem number	Multiplies the mass in kg by the distance.
B	Confusion set	$294 \text{ N} \times 10 \text{ m}$ ignores that the force and motion are perpendicular.
C	Adjacent category	Body effort is not physics work on the box.
D	Correct	No motion in the direction of the upward force.

5. Answer C Concept 4

A ~~jazz~~ radio station broadcasts at 100 MHz (1.0×10^8 Hz) ~~from a 150 m tower~~. Radio waves travel at 3.0×10^8 m/s. What is the wavelength?

Target word: wavelength **Decides:** $\lambda = v/f$.

A	Camouflage	The tower height does not set the wavelength.
B	Wrong stem number	Reports the wave speed.
C	Correct	$(3.0 \times 10^8) \div (1.0 \times 10^8)$.
D	Inverted operation	Multiplies v by f .

6. Answer A Concept 4 • transfer

A wave on a rope has a wavelength of 0.5 m and a speed of 2 m/s. What is its frequency?

Target word: solve for frequency **Decides:** $f = v/\lambda$.

A	Correct	$2 \div 0.5$.
B	Wrong stem number	Reports the speed.
C	Confusion set	$\lambda/v = 0.25$ s is the period, not the frequency.
D	Inverted operation	Multiplies v by λ .

7. Answer D Concept 5

A string of ~~warm white LED~~ lights ~~sold by a hardware store with free delivery~~ has a total resistance of 4 Ω and runs on a 12-V battery. What current flows?

Target word: current **Decides:** $I = V/R$.

A	Wrong stem number	Reports the voltage.
B	Confusion set	R/V flips the formula.
C	Inverted operation	Multiplies V by R .
D	Correct	$12 \div 4$.

8. Answer C Concept 5 • transfer

A 9-V battery drives 0.5 A through a small motor. What is the motor's resistance?

Target word: solve for resistance **Decides:** $R = V/I$.

A	Confusion set	I/V flips the formula.
B	Inverted operation	Multiplies V by I .
C	Correct	$9 \div 0.5$.
D	Wrong stem number	Reports the voltage.

9. Answer C Concept 6

A 200-g mug of water in a staff room microwave rated at 1,100 W is heated from 10°C to 35°C. Water's specific heat is 4.18 J/g·°C. How much heat does the water absorb?

Target word: heat absorbed **Decides:** $\Delta T = 25^\circ\text{C}$.

A	Wrong stem number	Uses the final temperature 35 instead of the change.
B	Dropped element	Leaves out ΔT .
C	Correct	$200 \times 4.18 \times 25$.
D	Camouflage	The microwave's power rating is not the heat absorbed.

10. Answer B Concept 7

Sodium, a metal, and chlorine, a nonmetal, combine to make table salt, which a food company sells in 1-kg bags. What kind of bond forms, and how?

Target word: bond type and mechanism **Decides:** Metal + nonmetal.

A	Inverted operation	Metals lose electrons; nonmetals gain them.
B	Correct	Na^+ and Cl^- attract.
C	Confusion set	Sharing occurs between nonmetals.
D	Adjacent category	Metallic bonding occurs among metal atoms.

11. Answer B Concept 8

Pond water near a closed paper mill tests at pH 5. Rainwater collected the same day at 18°C tests at pH 3. How does the rainwater's H^+ concentration compare with the pond's?

Target word: concentration ratio **Decides:** Two pH units lower means 10^2 times more H^+ .

A	Confusion set	Treats the scale as linear.
B	Correct	10×10 .
C	Wrong stem number	$5 \div 3$ divides the pH values.
D	Inverted operation	Lower pH means more H^+ , not less.

12. Answer A Concept 8 · transfer

Solution A has 1,000 times more H^+ than solution B, which has pH 9. What is the pH of solution A?

Target word: solve for pH **Decides:** $1,000 = 10^3$, so three units lower.

A	Correct	$9 - 3$.
B	Absolute language	pH can take any value along the scale.
C	Confusion set	Divides the pH by 3 instead of subtracting 3 units.
D	Inverted operation	More H^+ lowers pH; adding 3 raises it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Acceleration	Average acceleration = change in velocity ÷ time taken.	Multiplying by time or reporting the final speed.	A starting speed, an ending speed, and a time.
2	Conservation of momentum	In a collision with no outside force, total momentum (mass × velocity) before equals total after; objects that stick move with total momentum ÷ total mass.	Conserving speed instead of momentum, or leaving out one mass.	Carts or cars colliding and sticking together.
3	Work requires force along the motion	Work = force × distance moved in the direction of the force; a force perpendicular to the motion does no work.	Counting any effort or any force × distance as work.	Carrying something level at constant height.
4	Wave speed, frequency, and wavelength	$v = f\lambda$, so $\lambda = v/f$ and $f = v/\lambda$.	Multiplying when dividing is needed, or confusing frequency with period.	Two of speed, frequency, and wavelength given.
5	Ohm's law	$V = IR$, so $I = V/R$ and $R = V/I$.	Multiplying voltage by resistance to get current.	A voltage and a resistance or current.
6	Specific heat	Heat absorbed $q = m \times c \times \Delta T$, where ΔT is the change in temperature.	Using the final temperature instead of the change.	A mass, a specific heat, and two temperatures.
7	Ionic vs. covalent bonding	A metal and a nonmetal transfer electrons (metal to nonmetal) to form an ionic bond; nonmetals share electrons in covalent bonds.	Calling every compound covalent, or reversing electron flow.	Whether the elements are metals or nonmetals.
8	The pH scale is logarithmic	Each pH unit is a tenfold change in H^+ concentration; lower pH means more H^+ .	Treating pH as a linear scale.	Two pH values and "how many times...?"

Course scope (public description): This course provides students with an overview of the basic principles and unifying ideas of the physical sciences: physics, chemistry, and Earth sciences. Course materials focus on scientific reasoning and practical and everyday applications of physical science concepts to help students integrate conceptual knowledge with practical skills.